

HOXA13-Dependent Modulation of β -Catenin Transcriptional Efficacy Controls Spatial Stemness in a WNT-RA-HOX Model of Colonic Crypt Organization

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Abstract

Cross-regulation among WNT, retinoic acid (RA), and HOX signaling pathways governs the balance between stemness and differentiation in the colonic crypt and plays a central role in the initiation of colorectal cancer (CRC). Although canonical WNT/ β -catenin dynamics have been extensively modeled, the **quantitative** contribution of HOX-mediated transcriptional feedback to spatial patterning remains unclear. We develop a coupled system of nonlinear ordinary differential equations that integrates WNT/ β -catenin signaling, RA metabolism, and HOX gene regulation within a unified kinetic framework. The model includes two antagonistic HOX factors. HOXA5 promotes differentiation and suppresses WNT activity. HOXA13 enhances WNT-driven transcription and supports stem-like behavior. A key feature **of the model** is HOXA13-dependent modulation of β -catenin:TCF binding. This introduces state-dependent transcriptional efficacy rather than assuming fixed kinetics. The framework also incorporates MYC/MIZ1-mediated repression of HOXA5 together with RA-induced HOXA5 activation, forming a closed feedback loop between proliferation and differentiation. Steady-state solutions computed along a crypt-axis WNT gradient **??** reproduce physiological APC and WNT counter-gradients. These solutions

define a quantitative stemness index based on stabilized β -catenin, APC, and the HOXA13: β -catenin complex. APC reduction produces nonlinear amplification and spatial expansion of high-stemness regions. In addition, modulation of HOXA13-dependent transcriptional efficacy alone alters the extent of the proliferative **compartment zone of the crypt** without changing β -catenin stabilization. The model therefore identifies downstream transcriptional coupling as an independent regulator of stemness and provides a mechanistic explanation for crypt reprogramming during early tumorigenesis.

Keywords: Mathematical modeling; Nonlinear ordinary differential equations; Colorectal cancer; WNT/ β -catenin signaling; HOX gene regulation; Stemness dynamics

1 Introduction

Homeostasis of the colonic crypt relies on a finely regulated balance between stemness and differentiation, maintained through the coordinated action of multiple signaling pathways. Among these, canonical WNT/ β -catenin signaling, retinoic acid (RA) metabolism, and HOX gene regulation play central roles in controlling cell-fate decisions along the crypt axis [Clevers \(2006\)](#); [Reya and Clevers \(2005\)](#); [Pinto and Clevers \(2005\)](#). Disruption of this balance, most commonly through alterations in WNT signaling, can shift the system toward excessive proliferation and initiate colorectal cancer (CRC) development [Powell et al. \(1992\)](#); [Fodde et al. \(2001\)](#); [Klaus and Birchmeier \(2008\)](#). Although each of these pathways has been studied extensively, most existing mathematical models examine them in isolation and therefore do not capture the combined regulatory structure governing crypt-level organization. In particular, current WNT-based models typically assume a fixed relationship between stabilized β -catenin and transcriptional output, neglecting transcriptional modulation by downstream regulators. This limitation motivates the development of an integrated quantitative framework that connects WNT, RA, and HOX signaling within a single dynamical system. Mathematical modeling has long provided tools for studying complex biological regulation, including stochastic models of gene expression and multiscale representations of tumor growth and tissue organization [Swain et al. \(2002\)](#); [Watanabe et al. \(2016\)](#); [Rejniak and Anderson \(2011\)](#); [Macklin et al. \(2009\)](#).

Reaction-kinetic models of the WNT/ β -catenin pathway provide a well-established foundation for mathematical analysis of crypt dynamics. The seminal work of Lee et al. described APC- and Axin-mediated β -catenin degradation and demonstrated how scaffold availability regulates β -catenin stabilization and transcriptional output [Lee et al. \(2003\)](#). Subsequent extensions incorporated additional regulatory motifs and mechanistic structure to examine transcriptional dynamics, robustness, and oscillatory behavior [Wawra et al. \(2007\)](#); [van Leeuwen et al. \(2007\)](#); [Benary et al. \(2013\)](#). Other studies examined biochemical regulation of the pathway, including phosphorylation dynamics and compartmental transport of β -catenin, providing additional insight into how intracellular processes influence transcriptional activation [Sun \(2009\)](#); [Tan et al. \(2014\)](#). Complementary modeling efforts introduced compartmental representations,

including cytoplasmic–nuclear transport of β -catenin and its antagonists, showing how spatial separation of pathway components shapes signaling responses [Schmitz et al. \(2011, 2013\)](#). These molecular-scale descriptions were later embedded into crypt-level models, demonstrating that partial loss of APC elevates β -catenin activity and reproduces key features of early CRC progression [Emerick et al. \(2017, 2018\)](#). Additional theoretical studies have examined regulatory switches that link WNT signaling to epithelial-mesenchymal transition and downstream transcriptional responses in cancer systems [Gasior et al. \(2017\)](#). More recently, a reaction-kinetic model that incorporates cross-regulation between the WNT and RA pathways was developed to provide a more comprehensive description of signaling interactions relevant to CRC [Funk \(2025\)](#).

Together, these studies have advanced quantitative understanding of WNT regulation and its interaction with RA; however, they do not include HOX gene dynamics or HOX-mediated feedback on β -catenin signaling.

Mathematical modeling of RA signaling has developed along a related but largely separate trajectory. Ordinary differential equation models of RA metabolism have examined CYP26-mediated degradation, binding proteins, nuclear receptor activation, and the formation of stable RA gradients under developmental and physiological conditions [White et al. \(2007\)](#); [Ghaffari and Petzold \(2018\)](#); [Ghaffari et al. \(2019\)](#). Related developmental models have examined coupled signaling gradients, including interactions among RA, WNT, and FGF pathways in regulating differentiation zones [Joshi and Skromne \(2020\)](#). These models provide detailed representations of RA homeostasis and transcriptional control and have been extended to include mechanistic coupling between RA metabolism and WNT activity in CRC [Funk \(2025\)](#). Nevertheless, RA-focused models typically do not couple RA dynamics to HOX gene expression, nor do they account for HOX-dependent feedback on WNT regulation. This separation limits the ability to study how differentiation-promoting and stemness-sustaining signals interact within a unified quantitative framework.

Experimental evidence supports the integration of HOX gene regulation into models of WNT–RA cross-regulation. Earlier modeling studies also examined HOX gene regulation in developmental systems, demonstrating that stochastic regulatory mechanisms can reproduce spatial and temporal expression patterns during embryogenesis [Kastner et al. \(2002\)](#). HOX transcription factors regulate stem cell behavior and differentiation across development and cancer, and their dysregulation has been implicated in tumor initiation and progression [Bhatlekar et al. \(2018\)](#); [Shenoy et al. \(2022\)](#). In colorectal cancer, HOXA5 counteracts stem cell traits by inhibiting canonical WNT signaling and promoting differentiation [Ordóñez-Morán et al. \(2015\)](#); [Tan and Barker \(2015\)](#). In contrast, HOXA13 promotes colon cancer progression through β -catenin–dependent WNT signaling and is associated with enhanced transcriptional activation of WNT target genes [Gu et al. \(2020\)](#). Recent experimental evidence further indicates that HOXA5 and HOXA13 are regulated by both RA and WNT pathways in CRC-derived systems [Osmond \(2025\)](#). In particular, these studies demonstrate differential HOX responses to WNT activity and retinoid signaling in stem-like versus differentiated colorectal cancer cells. These findings provide direct biological justification for incorporating HOXA5- and HOXA13-mediated feedback into a unified mathematical description of pathway cross-regulation.

Retinoic acid signaling provides an additional differentiation-promoting axis that interacts with both WNT and HOX pathways. RA has long been implicated in cancer prevention and treatment through transcriptional control of differentiation programs [Connolly et al. \(2013\)](#); [Tang and Gudas \(2011\)](#). Importantly, RA availability is dynamically shaped by WNT-dependent regulation of RA metabolism, including induction of CYP26A1 in APC-deficient contexts [Shelton et al. \(2006\)](#). Recent CRC studies highlight CYP26A1 as a mechanistic link between WNT and RA signaling and as a potential target for differentiation-based therapeutic strategies [Facey et al. \(2024\)](#). These observations support modeling choices in which RA metabolism and its regulatory coupling to WNT are treated as dynamic components rather than prescribed inputs.

Despite substantial progress in modeling individual signaling pathways, no existing mathematical framework integrates WNT signaling, RA metabolism, and HOX gene regulation into a single system while allowing transcriptional output downstream of β -catenin to vary dynamically. As a result, current models cannot capture experimentally observed mechanisms in which HOX factors modulate WNT-driven stemness without directly altering upstream β -catenin stabilization [Ordóñez-Morán et al. \(2015\)](#); [Gu et al. \(2020\)](#). A coupled modeling approach is therefore needed to examine how feedback across these pathways maintains crypt organization under normal conditions and how its disruption contributes to early tumorigenesis.

In this work, we develop an integrated system of nonlinear ordinary differential equations that unifies WNT/ β -catenin signaling, RA metabolism, and HOX gene regulation within a single quantitative framework. The model extends existing formulations by explicitly incorporating HOXA5- and HOXA13-mediated transcriptional feedback, together with RA-dependent regulation of CYP26A1 [Emerick et al. \(2017\)](#); [Funk \(2025\)](#); [Facey et al. \(2024\)](#). This structure enables transcriptional efficacy downstream of stabilized β -catenin to emerge as a dynamic, state-dependent quantity rather than a fixed proportional output. We further introduce a quantitative measure of stemness derived from β -catenin signaling and HOX expression, allowing stemness to arise as a system-level outcome of the coupled dynamics. Through this approach, we examine how modulation of APC synthesis and HOX-dependent feedback reshapes crypt homeostasis and promotes stemness amplification, providing a mechanistic basis for understanding early CRC development.

2 Model formulation

We develop a coupled system of ordinary differential equations that integrates canonical WNT/ β -catenin signaling, retinoic acid (RA) metabolism, and HOX gene regulation to capture cross-regulation between stemness-promoting and differentiation-promoting programs in the colonic crypt. The WNT subsystem is adapted from the 17-variable reaction-kinetic framework of [Emerick et al. \(2017\)](#), which extends the classical Lee-Heinrich formulation [Lee et al. \(2003\)](#). The RA subsystem follows the retinoid signaling and metabolism model of [Funk \(2025\)](#). The HOX subsystem constitutes the main new contribution of this work and encodes experimentally supported regulatory roles of HOXA5 and HOXA13 in colorectal cancer (CRC) [Ordóñez-Morán](#)

enzymatic regulation are represented using saturating (Hill-type) rate laws. The three subsystems are coupled only through biologically motivated cross-regulatory terms, as summarized schematically in Fig. 1.

2.1 WNT/ β -catenin subsystem with HOX-dependent feedback

We retain the reaction structure, notation, and parameterization of [Emerick et al. \(2017\)](#), with state variables $X_1, \dots, X_{17}$ and reaction rates $r_1, \dots, r_{21}$ describing scaffold-mediated destruction complex dynamics and β -catenin stabilization. In this notation, X_{11} denotes stabilized (active) β -catenin, X_{14} denotes the β -catenin:TCF transcriptional complex, X_7 denotes APC, X_{12} denotes Axin, and $TCF^0 = X_{13} + X_{14}$ denotes the conserved total TCF [Lee et al. \(2003\)](#); [Emerick et al. \(2017\)](#). The full WNT system is listed in Appendix A; here we present only the components that are modified by or directly coupled to the HOX subsystem.

A central coupling mechanism is the modulation of β -catenin transcriptional efficacy by HOXA13. Following the rapid-equilibrium approximation commonly used in reduced WNT models [Mirams et al. \(2010\)](#); [Emerick et al. \(2017\)](#), we impose the binding relation

$$\frac{X_{11}X_{13}}{X_{14}} = \kappa(C_2), \quad (1)$$

where C_2 denotes the HOXA13: β -catenin complex and $\kappa(C_2)$ is a state-dependent dissociation constant. Using $TCF^0 = X_{13} + X_{14}$, this yields the algebraic closure

$$X_{14} = \frac{TCF^0 X_{11}}{\kappa(C_2) + X_{11}}. \quad (2)$$

An increase in C_2 decreases $\kappa(C_2)$, thereby increasing the fraction of β -catenin bound to TCF and enhancing WNT transcriptional output.

APC synthesis is redefined to incorporate repression by both WNT activity and HOXA13 in a form consistent with the regulatory structure used throughout the model. Specifically, reaction r_{18} is taken as

$$r_{18} : \text{APC synthesis} = \frac{v_{18}}{1 + \frac{X_{14}}{K_t} + \frac{X_{11}}{K_b} + \frac{H_2}{K_r}}, \quad (3)$$

where v_{18} is the maximal APC production rate, K_t and K_b are repression thresholds for β -catenin:TCF (X_{14}) and free β -catenin (X_{11}), respectively, and K_r is the repression threshold for HOXA13 (H_2). This rate law allows sustained WNT activity and elevated HOXA13 to jointly suppress APC production, consistent with the pro-tumorigenic role of HOXA13 in CRC [Gu et al. \(2020\)](#).

Under the same reduction assumptions used for the WNT subsystem, the coupled dynamics governing stabilized β -catenin and APC feedback can be written as

$$\frac{X_{11}}{K_8} \frac{dX_3}{dt} + \frac{X_{11}}{K_{17}} \frac{dX_7}{dt} + \left(1 + \frac{X_3}{K_8} + \frac{\kappa(C_2) TCF^0}{(\kappa(C_2) + X_{11})^2} + \frac{X_7}{K_{17}} + \frac{X_{12}}{K_{20}} \right) \frac{dX_{11}}{dt}$$

$$\begin{aligned}
& + \frac{X_{11}}{K_{20}} \frac{dX_{12}}{dt} = v_{12} + k_{-h10}C_2 \\
& - \left(k_{13} + \frac{k_9 X_3}{K_8} + k_{h10}H_2 + k_{h11}H_1 \right) X_{11}, \tag{4}
\end{aligned}$$

$$\begin{aligned}
& \left(1 + \frac{X_{11}}{K_8} \right) \frac{dX_3}{dt} + \left(1 + \frac{X_{12}}{K_7} + \frac{X_{11}}{K_{17}} \right) \frac{dX_7}{dt} + \left(\frac{X_3}{K_8} + \frac{X_7}{K_{17}} \right) \frac{dX_{11}}{dt} + \frac{X_7}{K_7} \frac{dX_{12}}{dt} \\
& = \frac{v_{18}}{1 + \frac{TCF^0 X_{11}}{K_t(\kappa(C_2) + X_{11})} + \frac{X_{11}}{K_b} + \frac{H_2}{K_r}} \\
& - k_{19}X_7 - \left(\frac{dX_4}{dt} + \frac{dX_9}{dt} \right). \tag{5}
\end{aligned}$$

Equations (4)–(28) identify the precise points at which HOX feedback acts on the WNT backbone: HOXA13 binds active β -catenin through $k_{h10}H_2X_{11}$, HOXA5 promotes β -catenin removal through $k_{h11}H_1X_{11}$, and HOXA13 represses APC synthesis through (3). All remaining WNT reactions and rate laws are unchanged from Emerick et al. (2017) and are listed in Appendix A.

2.2 HOX gene regulatory subsystem

The HOX regulatory subsystem constitutes the principal novel component of the model and captures cross-regulation between WNT transcriptional output, retinoic acid (RA) signaling, and HOX-mediated control of stemness and differentiation in colorectal tissue. We focus on two HOX genes with well-established but opposing roles in colorectal cancer: HOXA5, which promotes differentiation and suppresses WNT signaling, and HOXA13, which enhances WNT activity and supports stem-like behavior Ordóñez-Morán et al. (2015); Gu et al. (2020); Tan and Barker (2015). Experimental evidence indicates that these genes are embedded in feedback loops involving WNT-driven MYC expression, MYC–MIZ1-mediated transcriptional repression, and RA-dependent differentiation signals, motivating an explicit dynamical representation.

We model six HOX-related species: HOXA5 (H_1), HOXA13 (H_2), MIZ1 (M_1), MYC (M_2), the MYC:MIZ1 repressive complex (C_1), and the HOXA13: β -catenin complex (C_2). All species are treated as intracellular concentrations, and interactions are modeled using mass-action kinetics. Coupling to the WNT subsystem occurs through stabilized β -catenin (X_{11}) and the transcriptionally active β -catenin:TCF complex (X_{14}), while coupling to the RA subsystem occurs through the RA–RAR signal X_{8r} .

The HOX dynamics are governed by

$$\frac{dH_1}{dt} = v_{h1} - k_{h2}H_1, \tag{6}$$

$$\frac{dM_1}{dt} = v_{h3} - k_{h4}M_1 - k_{h5}M_2M_1 + k_{-h5}C_1, \tag{7}$$

$$\frac{dM_2}{dt} = v_{h6} - k_{h7}M_2 - k_{h5}M_2M_1 + k_{-h5}C_1, \quad (8)$$

$$\frac{dC_1}{dt} = k_{h5}M_2M_1 - k_{-h5}C_1, \quad (9)$$

$$\frac{dH_2}{dt} = v_{h8} - k_{h9}H_2 - k_{h10}H_2X_{11} + k_{-h10}C_2, \quad (10)$$

$$\frac{dC_2}{dt} = k_{h10}H_2X_{11} - k_{-h10}C_2. \quad (11)$$

HOXA5 synthesis is promoted by free MIZ1 and RA signaling and is indirectly suppressed by WNT through MYC–MIZ1 complex formation. This regulatory logic is encoded by

$$v_{h1} = v_{k1} + v_{k2}M_1 + \alpha_3 \frac{X_{8r}^{n_r}}{K_d + X_{8r}^{n_r}}, \quad (12)$$

where X_{8r} denotes the active RA–RAR complex. As MYC levels increase, MIZ1 is sequestered into C_1 , reducing the effective contribution of the M_1 term and thereby suppressing HOXA5 transcription, consistent with experimental observations [Ordóñez-Morán et al. \(2015\)](#); [Tan and Barker \(2015\)](#). In addition to transcriptional regulation, HOXA5 undergoes regulated degradation that depends on MYC–MIZ1 complex formation. We model the HOXA5 degradation rate as

$$k_{h2} = \frac{h_K + \alpha_1 C_1}{h_A + \phi C_1}, \quad (13)$$

where C_1 denotes the MYC–MIZ1 complex. This functional form ensures that increasing sequestration of MIZ1 into C_1 enhances HOXA5 degradation, providing a post-transcriptional mechanism through which elevated WNT signaling suppresses HOXA5. Together with the synthesis term in Eq. (12), this formulation captures both transcriptional and degradation-mediated repression of HOXA5 under MYC-dominant conditions.

MYC and HOXA13 are both modeled as WNT-responsive genes whose synthesis increases with transcriptionally active β -catenin:

$$v_{h6} = h_B + \alpha_2 \frac{X_{14}}{h_C + \beta X_{14}}, \quad (14)$$

$$v_{h8} = h_D + \alpha_4 \frac{X_{14}}{h_E + \beta' X_{14}}. \quad (15)$$

These forms ensure that elevated WNT signaling drives MYC expression, promoting MYC:MIZ1 complex formation and repression of HOXA5, while simultaneously inducing HOXA13. HOXA13 further enhances WNT signaling by binding stabilized β -catenin and modulating β -catenin:TCF affinity through Eqs. (1)–(2), closing a positive feedback loop.

Through these interactions, the HOX subsystem integrates differentiation-promoting RA signals and stemness-promoting WNT signals into a unified regulatory structure. High RA availability favors HOXA5 expression and WNT suppression,

whereas strong WNT activity elevates MYC and HOXA13, represses HOXA5, and stabilizes a stem-like transcriptional state. This balance is central to crypt homeostasis and is disrupted under oncogenic perturbations such as APC loss.

2.3 Retinoic acid subsystem and CYP-mediated cross-regulation

The retinoic acid (RA) subsystem captures intracellular retinoid metabolism, receptor-mediated transcriptional signaling, and enzymatic degradation following the framework of Funk (2025). RA acts as a differentiation-promoting signal in the intestinal epithelium and counterbalances WNT-driven stemness by activating nuclear retinoic acid receptors (RARs), which regulate transcription of downstream targets including HOX genes Tang and Gudas (2011); Connolly et al. (2013). In the present model, RA signaling is coupled to both the HOX and WNT subsystems through HOXA5 induction and CYP26A1-mediated RA clearance.

The RA subsystem consists of thirteen state variables ($X_{1r}, \dots, X_{13r}$) describing retinol uptake, intracellular conversion, RAR binding, and CYP-mediated metabolism. The full system of RA equations is provided in Appendix B to avoid repetition. Here we emphasize only the regulatory components required for cross-regulation with the WNT and HOX subsystems.

The active RA–RAR transcriptional signal is represented by X_{8r} , which enters the HOXA5 synthesis rate in Eq. (12). Through this coupling, increased RA signaling promotes HOXA5 expression and thereby indirectly suppresses WNT activity via enhanced β -catenin removal in Eq. (4). The availability of X_{8r} is, in turn, regulated by CYP26A1, an RA-degrading enzyme whose expression is dynamically controlled within the RA subsystem.

In the original funk2025modeling model, CYP synthesis depends only on intracellular retinoid concentration and is given by the saturating rate law

$$v_{19} = \frac{v_{18r} + MX_{4r}^2}{K + X_{4r}^2}, \quad (16)$$

where X_{4r} denotes the intracellular retinoid species driving CYP induction, v_{18r} is the basal CYP production rate, M sets the maximal inducible contribution, and K is a half-saturation constant Funk (2025). While this formulation captures RA-dependent feedback, it does not account for experimental evidence that CYP26A1 is also transcriptionally upregulated by WNT/ β -catenin signaling Shelton et al. (2006); Facey et al. (2024).

To incorporate this additional layer of cross-regulation, we redefine the CYP synthesis rate as

$$v_{19}^{\text{eff}} = \frac{\nu_{18} + \mu X_{4r}^2}{K + X_{4r}^2} + \frac{\delta X_{11}}{(K + X_{4r}^2)(\kappa(C_2) + X_{11})}. \quad (17)$$

The first term retains the original RA-dependent saturating response, where $\nu_{18} > 0$ denotes basal CYP synthesis and $\mu > 0$ sets the maximal RA-induced contribution.

The second term introduces a WNT-dependent component proportional to stabilized β -catenin X_{11} , scaled by $\delta > 0$, and modulated by the transcriptional efficacy factor $(\kappa(C_2) + X_{11})^{-1}$. This factor couples CYP induction to the HOXA13-dependent modulation of β -catenin:TCF binding encoded in Eqs. (1)–(2), ensuring that WNT-driven CYP upregulation is strongest when transcriptional output is high.

Through Eq. (17), elevated WNT signaling promotes CYP26A1 expression, accelerating RA degradation and reducing the RA–RAR signal X_{8r} . This suppresses HOXA5 induction and further reinforces WNT activity, establishing an indirect feedback loop in which WNT signaling regulates RA availability, which in turn modulates HOX-dependent differentiation control. Conversely, under conditions of low WNT activity, reduced CYP expression allows RA levels to rise, strengthening HOXA5-mediated differentiation signals and attenuating β -catenin stabilization. In this way, CYP26A1 serves as a critical regulatory interface linking WNT transcriptional output, RA availability, and HOX-dependent control of stemness.

2.4 Nondimensionalization and parameter selection

Before numerical analysis, the coupled WNT–RA–HOX system is nondimensionalized to reduce parameter redundancy, clarify dominant regulatory balances, and place all variables on comparable temporal and concentration scales [Lee et al. \(2003\)](#); [Emerick et al. \(2017\)](#). This formulation also facilitates consistent comparison across spatial positions along the crypt axis.

Time and concentration scaling.

Time is rescaled using the characteristic turnover rate of the active β -catenin destruction complex, which sets the fastest intrinsic timescale of WNT signaling [Lee et al. \(2003\)](#); [Emerick et al. \(2017\)](#). Let k_5 denote the dissociation or inactivation rate of this complex (reaction r_5 in the WNT subsystem). We define the dimensionless time

$$\tau = k_5 t,$$

so that one unit of τ corresponds to a single β -catenin turnover cycle. With $k_5^{-1} \approx 7.5$ minutes, this choice captures the rapid WNT response while remaining compatible with slower transcriptional and metabolic feedback processes [Mirams et al. \(2010\)](#).

All concentrations are rescaled by a common reference level σ , with

$$x_i = \frac{X_i}{\sigma}.$$

Guided by steady-state analysis of the RA subsystem and reported intracellular retinoid concentrations, we take $\sigma = 1000$ nM [White et al. \(2007\)](#); [Ghaffari et al. \(2019\)](#). Under this scaling, RA-related variables are $\mathcal{O}(1)$, while WNT and HOX components typically lie in the 10^{-2} – 10^{-1} range under physiological conditions. This choice improves numerical conditioning and avoids artificial scale separation. All threshold constants and dissociation parameters are normalized consistently by σ .

Representative dimensionless couplings.

After rescaling, the system can be written compactly as

$$\frac{d\mathbf{z}}{d\tau} = \mathbf{F}(\mathbf{z}; \boldsymbol{\theta}),$$

where $\mathbf{z}$ collects all dimensionless state variables and $\boldsymbol{\theta}$ denotes the dimensionless parameter set. Rather than reproducing the full system here, we highlight the key cross-regulatory relations that couple the WNT, HOX, and RA pathways.

The HOXA13-modulated rapid-equilibrium relation for β -catenin:TCF binding becomes

$$B_t = \frac{TCF^0 B_a}{\kappa_d(C_i) + B_a}, \quad (18)$$

where B_a denotes stabilized β -catenin, B_t the β -catenin:TCF transcriptional complex, TCF^0 the conserved total TCF level, and $\kappa_d(C_i)$ the HOXA13-dependent dissociation function controlled by the HOXA13: β -catenin complex C_i [Gu et al. \(2020\)](#).

APC synthesis rescales to

$$\hat{r}_{18} = \frac{\hat{v}_{18}}{1 + \frac{B_t}{k_t} + \frac{B_a}{k_b} + \frac{H_i}{k_r}}, \quad (19)$$

making explicit how WNT transcriptional output and HOXA13 jointly repress APC production in dimensionless form, consistent with APC–WNT antagonism and HOX-mediated feedback observed in colorectal cancer [Lee et al. \(2003\)](#); [Emerick et al. \(2017\)](#); [Gu et al. \(2020\)](#).

Finally, the CYP26A1 synthesis term becomes

$$\hat{v}_{19}^{\text{eff}} = \frac{\hat{v}_{18} + \hat{\mu} R^2}{1 + R^2/k} + \frac{\hat{\delta} B_a}{(1 + R^2/k) (\kappa_d(C_i) + B_a)}, \quad (20)$$

where the first contribution represents RA-dependent induction, while the second captures an intrinsic WNT-dependent contribution mediated by stabilized β -catenin B_a and modulated by HOXA13 through the dissociation function $\kappa_d(C_i)$ [Funk \(2025\)](#); [Facey et al. \(2024\)](#). All remaining equations follow analogously under the same scaling and are provided in the appendices.

Parameter selection.

Model parameters were drawn from a combination of established kinetic models and recent experimental evidence, with the goal of preserving validated pathway behavior while enabling investigation of HOX-mediated regulatory effects. Parameters associated with the core WNT signaling subsystem were adopted directly from the reaction–kinetic frameworks of [Lee et al. \(2003\)](#) and [Emerick et al. \(2017\)](#). These values reproduce experimentally consistent timescales for β -catenin turnover, APC-mediated regulation, and β -catenin:TCF binding, ensuring that baseline WNT dynamics align with prior modeling studies.

Parameters governing retinoic acid (RA) metabolism and signaling were taken from the model of Funk (2025), which captures intracellular retinoid transport, receptor binding, and CYP26A1-mediated degradation. These parameters yield RA half-lives on the order of hours and high-affinity RA–RAR interactions, consistent with biochemical measurements and previous RA-focused modeling work.

The HOX subsystem constitutes the principal novel component of the model. Parameter values governing HOXA5 and HOXA13 regulation were selected to reflect experimentally supported qualitative behavior rather than to fit specific quantitative datasets, which are currently unavailable for these transcriptional feedbacks. In particular, HOXA5-associated parameters were chosen so that increased HOXA5 expression enhances β -catenin removal and suppresses WNT transcriptional output, while HOXA13-associated parameters promote stabilization of nuclear β -catenin and amplification of transcriptional efficacy. These choices are consistent with experimental findings demonstrating that HOXA5 enforces differentiation programs, whereas HOXA13 reinforces WNT-driven stemness in colorectal cancer systems Ordóñez-Morán et al. (2015); Tan and Barker (2015); Gu et al. (2020); Bhatlekar et al. (2018).

Several HOX-related parameter ranges were further informed by recent experimental studies examining HOX-dependent modulation of WNT activity in colorectal cancer models Osmond (2025). In particular, the strength of HOXA13-mediated enhancement of β -catenin transcriptional output and the sensitivity of HOXA5 induction to RA signaling were constrained to remain consistent with observed perturbation responses reported therein. While these data do not directly specify kinetic rate constants, they restrict biologically plausible regimes and strengthen the experimental grounding of the HOX regulatory structure.

All threshold parameters were chosen so that moderate changes in HOX expression produce substantial regulatory effects, reflecting the strong transcriptional influence observed experimentally. Parameter sets were further constrained to ensure bounded solutions and a biologically reasonable baseline steady state corresponding to normal crypt homeostasis. Sensitivity analyses confirmed that the qualitative behaviors reported in this work—particularly the expansion of stem-like states under APC loss—are robust across a broad range of HOX parameter values, provided the opposing roles of HOXA5 and HOXA13 are maintained. Complete parameter definitions, nondimensional values, and literature sources are provided in the appendices.

2.5 Definition of stemness

To quantify the balance between stem-like and differentiated states along the colonic crypt, we define a dimensionless stemness index derived directly from the steady-state solution of the coupled WNT–RA–HOX system. Stemness here reflects sustained WNT transcriptional activity, reduced differentiation pressure from APC, and reinforcement of β -catenin signaling by HOXA13.

Following the conceptual framework introduced by Emerick et al. (2017), stemness is assumed to increase with stabilized β -catenin and decrease with APC abundance. We generalize this definition by incorporating HOXA13-mediated modulation

of β -catenin:TCF binding. Specifically, the effective dissociation constant between β -catenin and TCF is defined as a Hill-type function of the HOXA13: β -catenin complex concentration C_i ,

$$\kappa(C_i) = \gamma_0 + \frac{\gamma_1 C_i^{n_c}}{C_s^{n_c} + C_i^{n_c}}, \quad n_c = 2, \quad (21)$$

which increases monotonically with C_i and saturates at $\gamma_0 + \gamma_1$. This formulation captures the experimentally supported role of HOXA13 in enhancing β -catenin nuclear signaling.

Using this effective binding rate, the stemness index at crypt position s is defined as

$$\sigma_s = \frac{B_a \sqrt{TCF^0}}{P \sqrt{\kappa(C_i) + B_a}}, \quad (22)$$

where B_a and P denote the steady-state dimensionless concentrations of stabilized β -catenin and APC, respectively, and TCF^0 is the conserved total TCF level. By construction, σ_s increases with WNT transcriptional output and HOXA13 reinforcement, and decreases with APC-mediated differentiation pressure.

In the absence of HOXA13 regulation (i.e., $\gamma_1 = 0$), $\kappa(C_i)$ reduces to a constant and (22) recovers exactly the stemness definition of Emerick et al. (2017). Thus, σ_s provides a strict generalization of prior formulations while preserving their limiting behavior.

The stemness index is evaluated at steady state for each crypt position after convergence of the ODE system under the local WNT input. Under physiological conditions, σ_s is highest near the crypt base and decreases toward the lumen, reproducing the known spatial organization of stem and differentiated compartments. Deviations from this profile under APC loss or altered HOX regulation provide a quantitative measure of crypt reprogramming, which we analyze in the following section.

3 Numerical results

3.1 Baseline crypt organization under normal APC conditions

Figure 2 shows the steady-state spatial profiles of key WNT and HOX pathway components along the colonic crypt under baseline (wild-type) APC synthesis. Steady states were obtained by imposing a spatially decaying WNT input and integrating the coupled WNT–HOX–RA system to equilibrium independently at each crypt position.

Under baseline conditions, APC concentration $P(s)$ increases monotonically from the crypt base toward the lumen (Fig. 2a), establishing a spatial gradient that constrains WNT activity. In contrast, stabilized β -catenin $B_a(s)$ is maximal at the crypt base and decreases toward the upper crypt (Fig. 2b). The resulting β -catenin:TCF transcriptional output $B_t(s)$ closely follows this pattern (Fig. 2c), confining strong WNT-driven transcription to the lower crypt.

The HOXA13: β -catenin complex $C_i(s)$ exhibits a similar spatial enrichment in the stem and transit-amplifying regions and declines toward the differentiated compartment (Fig. 2d). This distribution reflects the dependence of C_i on stabilized β -catenin and establishes a spatially varying influence on transcriptional efficacy through the

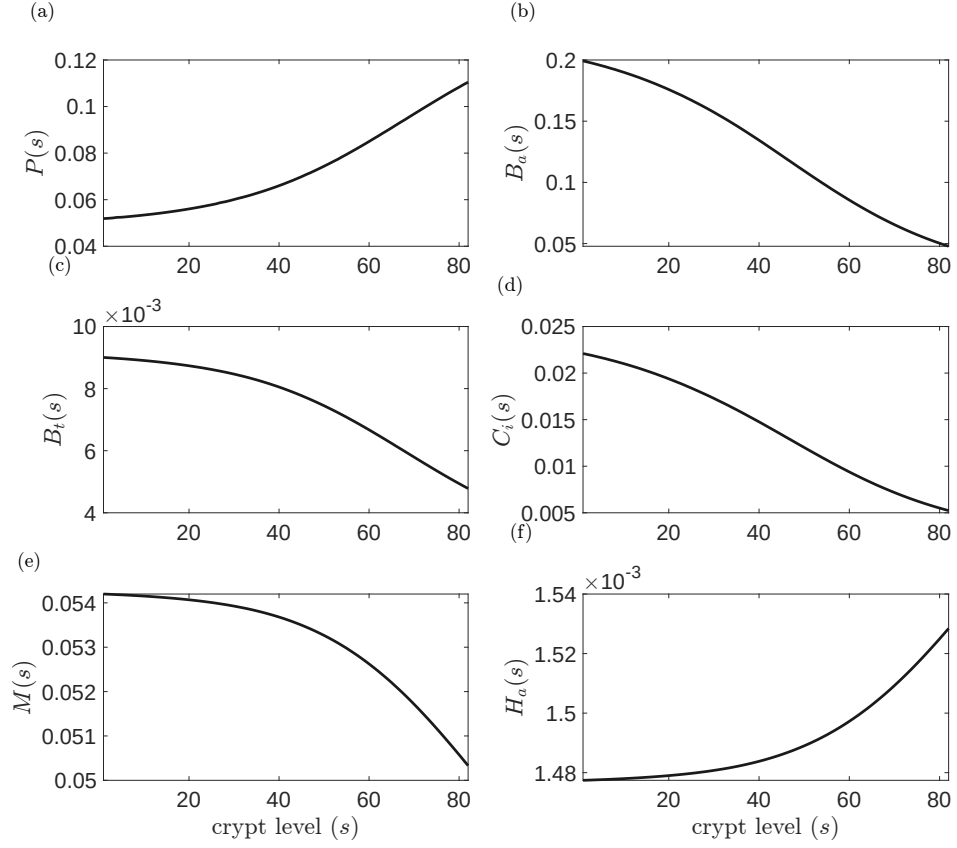


Fig. 2: Baseline spatial organization of WNT and HOX pathway components along the crypt axis. (a) APC concentration $P(s)$. (b) Active β -catenin $B_a(s)$. (c) β -catenin:TCF transcriptional output $B_t(s)$. (d) HOXA13: β -catenin complex $C_i(s)$. (e) MYC expression $M(s)$. (f) HOXA5 expression $H_a(s)$.

dynamic β -catenin:TCF binding rate. MYC expression $M(s)$ mirrors the transcriptional output profile (Fig. 2e), whereas HOXA5 expression $H_a(s)$ displays an opposing gradient, with minimal levels at the crypt base and maximal expression toward the lumen (Fig. 2f).

Together, these profiles define a reference crypt organization characterized by a proliferative lower compartment with high WNT activity and a differentiated upper compartment with elevated APC and HOXA5. This baseline state provides a quantitative reference for assessing the effects of altered transcriptional regulation and APC perturbations.

3.2 Dynamic β -catenin:TCF binding reshapes pathway dynamics

We next examined how HOXA13-dependent regulation of the β -catenin:TCF binding rate alters pathway dynamics. Figure 3 compares time-course simulations under identical WNT stimulation for a constant binding parameter K_{16} and for a dynamically regulated $K_{16}(C_i)$.

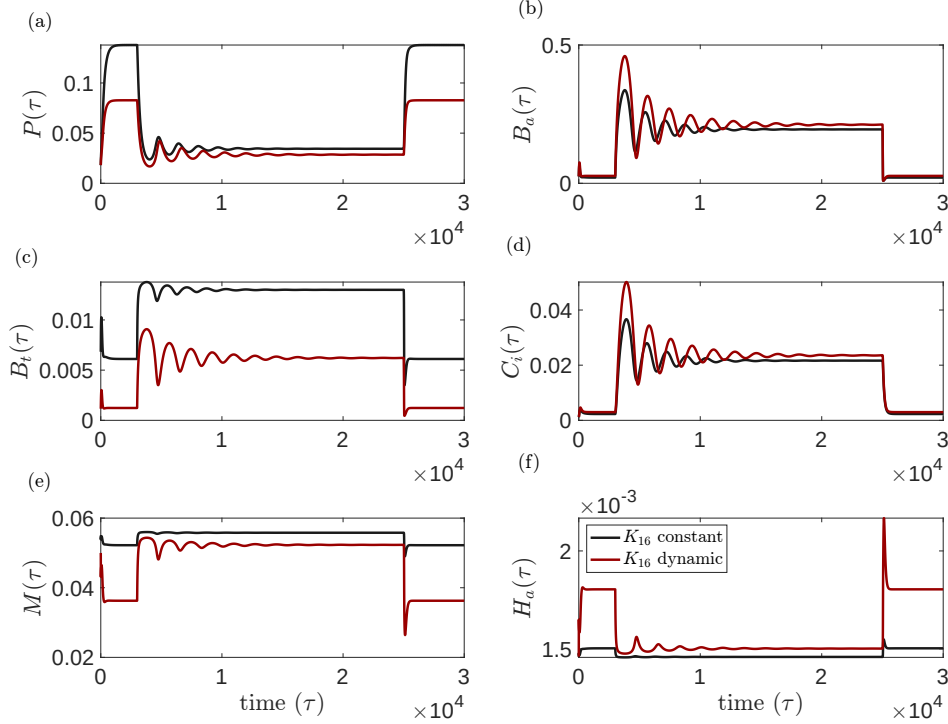


Fig. 3: Time-course comparison of constant versus dynamic K_{16} regulation. (a) APC $P(\tau)$. (b) Active β -catenin $B_a(\tau)$. (c) β -catenin:TCF output $B_t(\tau)$. (d) HOXA13: β -catenin complex $C_i(\tau)$. (e) MYC $M(\tau)$. (f) HOXA5 $H_a(\tau)$. Black curves correspond to constant K_{16} , red curves to HOXA13-dependent $K_{16}(C_i)$.

APC and stabilized β -catenin exhibit nearly identical trajectories in both formulations (Fig. 3a–b), indicating that dynamic regulation of K_{16} does not alter upstream WNT signal propagation or β -catenin stabilization kinetics. In both cases, β -catenin accumulates rapidly and reaches comparable steady-state levels.

In contrast, pronounced differences emerge at the transcriptional level. When K_{16} is dynamically regulated, the β -catenin:TCF transcriptional output $B_t(\tau)$ is substantially attenuated relative to the constant- K_{16} case (Fig. 3c). This demonstrates a decoupling between β -catenin abundance and transcriptional efficacy: stabilized β -catenin alone is no longer sufficient to drive maximal transcriptional output.

This attenuation propagates downstream. MYC expression is reduced under dynamic K_{16} regulation (Fig. 3e), reflecting its dependence on sustained transcriptional activation. At the same time, the HOXA13: β -catenin complex $C_i(\tau)$ transiently increases (Fig. 3d), reinforcing the feedback loop that governs transcriptional efficacy. HOXA5 expression remains suppressed throughout the simulation window (Fig. 3f), indicating that the overall balance of signaling remains biased toward a WNT-dominant state.

These results show that HOXA13-dependent modulation of β -catenin:TCF binding selectively reshapes downstream pathway dynamics by tuning transcriptional efficiency, while leaving upstream WNT activation intact.

3.3 APC synthesis reduction amplifies and redistributes WNT signaling

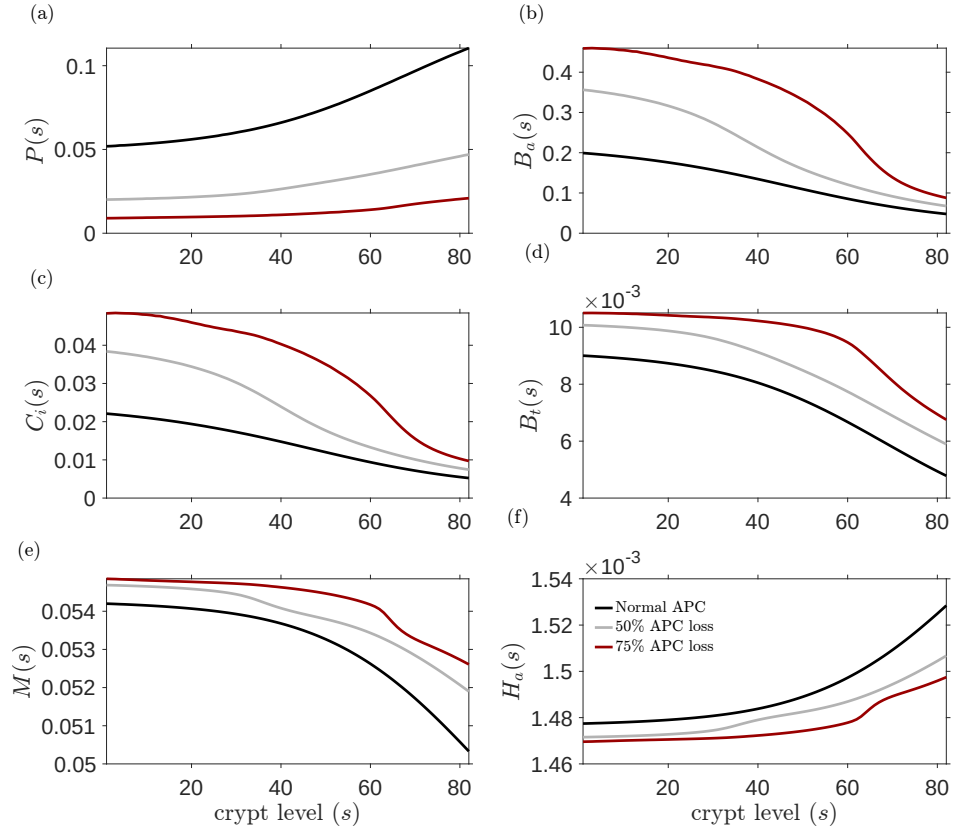


Fig. 4: Spatial profiles under graded APC synthesis reduction. (a) APC $P(s)$. (b) Active β -catenin $B_a(s)$. (c) HOXA13: β -catenin complex $C_i(s)$. (d) β -catenin:TCF output $B_t(s)$. (e) MYC $M(s)$. (f) HOXA5 $H_a(s)$.

To investigate the impact of APC loss, we simulated graded reductions in the APC synthesis rate and computed steady-state spatial profiles along the crypt axis.

Reducing APC synthesis lowers $P(s)$ uniformly across the crypt (Fig. 4a), weakening the spatial constraint on WNT signaling. As a result, stabilized β -catenin $B_a(s)$ increases at all crypt positions (Fig. 4b), with the largest absolute increases occurring near the crypt base.

The increased availability of stabilized β -catenin enhances formation of the HOXA13: β -catenin complex $C_i(s)$ (Fig. 4c), strengthening the transcriptional feedback encoded by $K_{16}(C_i)$. Consequently, the β -catenin:TCF output $B_t(s)$ exhibits both increased magnitude and expanded spatial extent (Fig. 4d), extending upward into regions that previously displayed low transcriptional activity.

Downstream responses reflect this redistribution. MYC expression increases throughout the crypt (Fig. 4e), while HOXA5 expression is progressively suppressed, particularly in regions that previously exhibited strong differentiation-associated signaling (Fig. 4f). These changes indicate that APC reduction induces both amplification and spatial redistribution of WNT-driven transcription through HOXA13-dependent feedback.

3.4 APC loss expands the stem compartment through HOXA13– β -catenin coupling

Under baseline conditions, the stemness index $\sigma(s)$ is maximal at the crypt base and decreases monotonically toward the lumen (Fig. 5a). Progressive APC depletion elevates $\sigma(s)$ throughout the crypt, but the response is not a uniform amplification. Instead, APC loss produces a spatial redistribution of stemness characterized by an upward displacement of high- σ regions.

Applying a fixed stemness threshold σ_{th} highlights this reorganization explicitly (Fig. 5b). While the threshold boundary is confined to the lower crypt under normal APC synthesis, APC reduction shifts this boundary upward in a nonlinear manner, indicating expansion of the stem and transit-amplifying compartments.

These results demonstrate that APC loss reorganizes crypt architecture by coupling elevated stabilized β -catenin with HOXA13-dependent enhancement of transcriptional efficacy. The resulting expansion of stem-like states provides a quantitative measure of crypt reprogramming driven by feedback-mediated pathway coupling.

To quantify how APC loss alters crypt organization, we evaluated the stemness index $\sigma(s)$ along the crypt axis under graded APC reduction.

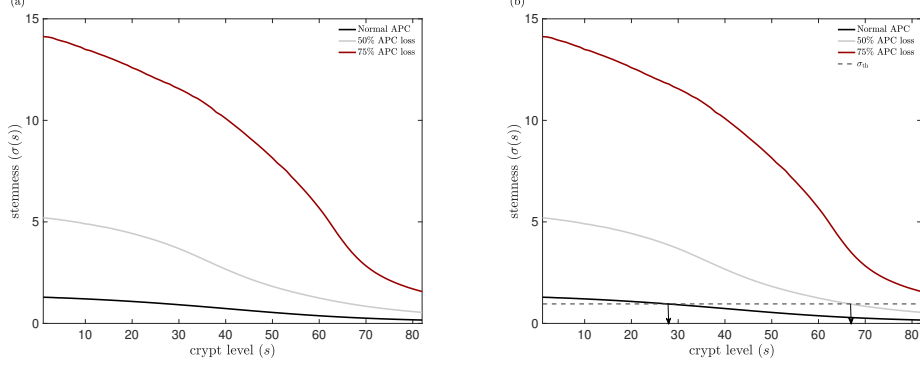


Fig. 5: Stemness profiles under graded APC loss. (a) $\sigma(s)$ for normal APC, 50% APC loss, and 75% APC loss. (b) Same profiles with stemness threshold σ_{th} .

3.5 Stemness modulation through transcriptional efficacy

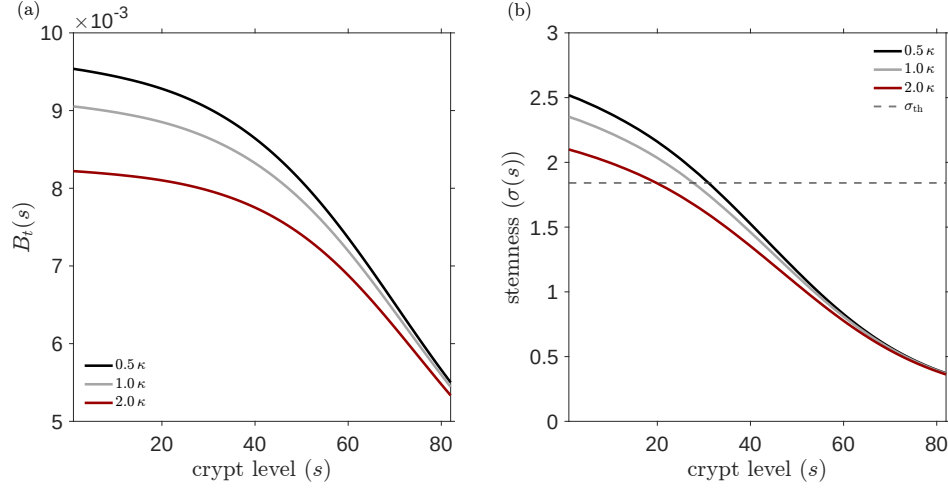


Fig. 6: Effect of transcriptional efficacy modulation on WNT output and stemness. (a) Steady-state spatial profiles of β -catenin:TCF transcriptional output $B_t(s)$ for scaled values of the parameter κ , which rescales the HOXA13-dependent dissociation rate K_{16} . (b) Corresponding stemness profiles $\sigma(s)$ computed from the model-defined proxy, with the stemness threshold σ_{th} (dashed line) indicating the approximate boundary of the proliferative compartment at $\sim 33\%$ of the crypt. All simulations are performed under normal APC and WNT conditions.

Although APC loss is a primary driver of elevated WNT signaling and stemness, the model also allows stem-associated behavior to be regulated downstream of β -catenin stabilization. To examine this effect independently, we modulated the

parameter κ , which rescales the HOXA13-dependent dissociation rate K_{16} governing β -catenin:TCF transcriptional activity, while keeping APC and WNT inputs fixed.

Figure 6(a) shows the steady-state spatial profiles of transcriptional output $B_t(s)$ under graded values of κ . Decreasing κ increases $B_t(s)$ throughout the crypt, whereas increasing κ uniformly suppresses transcriptional output. The effect is most pronounced in the lower crypt, where β -catenin signaling is strongest, indicating that modulation of K_{16} directly controls the efficiency of WNT-driven transcription rather than upstream signal availability.

The corresponding stemness profiles are shown in Figure 6(b). Variation in κ produces systematic shifts in $\sigma(s)$ without altering the qualitative shape of the spatial profile. Lower κ values expand the region in which stemness exceeds the threshold σ_{th} , defined at approximately the lower third of the crypt, while higher κ values contract this region and reduce overall stemness levels. These changes arise solely from altered transcriptional efficacy and do not depend on modifications to APC-mediated β -catenin degradation.

In all, these results demonstrate that stemness in the model is sensitive not only to upstream WNT regulation but also to downstream transcriptional control mediated by HOXA13-dependent modulation of β -catenin:TCF interactions. This provides a complementary mechanism through which stem-associated signaling can be amplified or constrained, even in the absence of additional genetic perturbations.

4 Discussion

In this work, we developed a coupled mathematical model integrating WNT/ β -catenin signaling, retinoic acid (RA) metabolism, and HOX gene regulation to investigate how transcriptional feedback and pathway cross-regulation govern spatial organization and stemness along the colonic crypt. By extending established kinetic models of WNT and RA signaling with a mechanistically motivated HOX regulatory layer, the model captures how intracellular transcriptional context modulates pathway output and reshapes crypt-level behavior under both normal and perturbed conditions.

A central finding of this study is that transcriptional output downstream of WNT signaling need not scale directly with stabilized β -catenin levels. Classical WNT models typically assume that once β -catenin is stabilized and enters the nucleus, its binding to TCF and subsequent transcriptional activation proceed with fixed kinetics [Lee et al. \(2003\)](#); [Mirams et al. \(2010\)](#); [Emerick et al. \(2017\)](#). Our results demonstrate that introducing HOXA13-dependent modulation of the β -catenin:TCF binding interaction fundamentally alters this paradigm. In the model, upstream WNT activation and β -catenin accumulation remain largely unchanged, yet transcriptional efficacy becomes dynamically regulated through the formation of the HOXA13: β -catenin complex. This decoupling between signal stabilization and transcriptional output provides a mechanistic explanation for experimental observations in which elevated β -catenin does not always produce proportionally increased WNT target gene expression [Ordóñez-Morán et al. \(2015\)](#); [Gu et al. \(2020\)](#).

The numerical results show that this dynamic transcriptional regulation reshapes downstream signaling without disrupting core WNT pathway dynamics. When the

β -catenin:TCF binding rate is allowed to depend on HOXA13, transcriptional output becomes self-limiting despite sustained β -catenin stabilization. As a consequence, MYC activation is attenuated relative to classical WNT models, while HOXA13-mediated feedback is reinforced. This behavior highlights the role of HOXA13 as an intracellular modulator that tunes WNT output rather than acting as a simple on/off regulator. Such a mechanism is consistent with experimental studies reporting that HOXA13 interacts with nuclear β -catenin and enhances transcriptional control in colorectal cancer cells [Gu et al. \(2020\)](#); [Bhatlekar et al. \(2018\)](#), suggesting that transcriptional regulation downstream of WNT is highly context dependent. This transcriptional modulation is also consistent with recent experimental perturbation studies demonstrating that HOXA13 alters WNT-driven transcriptional responses and stem-associated phenotypes in colorectal cancer systems [Osmond \(2025\)](#).

The model further reveals how this HOXA13-dependent feedback becomes particularly impactful under APC loss, a defining event in early colorectal tumorigenesis. Reductions in APC synthesis increase stabilized β -catenin across the crypt, as expected, but also amplify formation of the HOXA13: β -catenin complex. This amplification strengthens transcriptional feedback and causes β -catenin:TCF activity to spread upward along the crypt axis, extending well beyond the stem cell niche. Importantly, this expansion is not merely a uniform increase in signaling magnitude but a spatial redistribution of transcriptional activity. Regions that normally support differentiation begin to exhibit sustained WNT transcriptional output, consistent with experimental observations of expanded proliferative compartments in APC-mutant crypts [Vermeulen et al. \(2010\)](#); [Sansom et al. \(2004\)](#).

Concomitantly, APC loss suppresses differentiation-promoting signals mediated by HOXA5. HOXA5 expression is progressively reduced across the crypt as WNT/MYC signaling intensifies, particularly through MYC-MIZ1-mediated repression [Ordóñez-Morán et al. \(2015\)](#); [Tan and Barker \(2015\)](#). The combined effect of increased WNT transcriptional output and reduced HOXA5 activity produces a coordinated shift toward a stem-like transcriptional program throughout the tissue. In this sense, APC loss does not simply deregulate WNT signaling but actively reorganizes crypt identity by engaging HOX-dependent transcriptional feedback loops.

To quantify this reorganization, we introduced a generalized stemness index that extends prior formulations by incorporating HOXA13-dependent modulation of transcriptional efficacy. While earlier models defined stemness primarily through the antagonism between β -catenin and APC [Emerick et al. \(2017\)](#), our formulation allows stemness to depend on both signaling strength and transcriptional context. The resulting stemness profiles reveal that APC loss induces a nonlinear expansion of high-stemness regions along the crypt, rather than a simple amplification of stemness at the base. This distinction is biologically important, as experimental studies suggest that tumor initiation involves upward expansion of stem and transit-amplifying compartments rather than uniform increases in proliferation [Schepers et al. \(2012\)](#); [Vermeulen et al. \(2013\)](#).

Importantly, the model also indicates that stemness can be modulated independently of APC perturbation through changes in transcriptional efficacy. When the

HOXA13-dependent scaling parameter κ is varied under fixed WNT input and normal APC levels, both the magnitude of β -catenin:TCF transcriptional output and the spatial extent of elevated stemness shift systematically. These changes occur without altering upstream β -catenin stabilization dynamics, demonstrating that downstream modulation of transcriptional coupling alone is sufficient to expand or contract the proliferative compartment in the model. This behavior underscores that stemness is governed not only by pathway activation at the level of signal stabilization, but also by intracellular regulation of transcriptional efficiency.

The integration of RA signaling further strengthens the model’s biological relevance. RA is a well-established driver of intestinal differentiation and a regulator of HOX gene expression [Tang and Gudas \(2011\)](#); [Connolly et al. \(2013\)](#). In the present framework, RA promotes HOXA5 expression while being indirectly suppressed by WNT-driven CYP26A1 activity. This coupling provides a mechanistic basis for the observed segregation between WNT-rich, RA-poor stem zones and RA-rich differentiated regions [White et al. \(2007\)](#); [Ghaffari et al. \(2019\)](#). By linking RA availability to WNT and HOX activity, the model captures how differentiation signals are progressively silenced as WNT and HOXA13 feedback intensify during tumor initiation.

Despite its mechanistic scope, the model has limitations that point to future directions. Precise kinetic parameters for HOX transcriptional regulation remain scarce, and parameter values were therefore chosen to reproduce qualitative biological behavior rather than fit quantitative datasets. As experimental measurements of HOX dynamics become available, model calibration and validation will be essential. In addition, the present framework treats each crypt position independently and does not include explicit cell migration or lineage dynamics, which are known to influence crypt maintenance [Itzkovitz et al. \(2012\)](#). Coupling the signaling model to spatial or agent-based descriptions of cell movement would enable investigation of how signaling-driven stemness translates into long-term tissue architecture.

Conclusion

In conclusion, this study demonstrates that transcriptional regulation downstream of WNT signaling plays a decisive role in shaping stemness and spatial organization in the colonic crypt. By introducing HOXA13-dependent modulation of β -catenin:TCF binding, the model reveals how HOX dysregulation can amplify WNT signaling effects without altering upstream pathway activation. Under APC loss, this mechanism drives expansion of stem-like states and suppression of differentiation cues, providing a mechanistic link between APC mutations, HOX imbalance, and early crypt reprogramming in colorectal cancer. More broadly, these results highlight the importance of modeling transcriptional efficacy as a dynamic, state-dependent process, offering a framework for studying how intracellular regulatory networks reshape canonical signaling pathways during disease progression.

A Supplementary variables and parameters

A.1 Model variables and biological interpretation

Table 1 summarizes the key state variables used in the coupled model.

Table 1: Model variables and biological interpretation.

Variable	Description
X_{11}	Stabilized (active) β -catenin
X_{14}	β -catenin:TCF transcriptional complex
X_7	APC tumor suppressor
X_{12}	Axin scaffold protein
H_1	HOXA5 (differentiation-promoting HOX factor)
H_2	HOXA13 (stemness-promoting HOX factor)
M_2	MYC oncogene
M_1	MIZ1 transcription factor
C_1	MYC:MIZ1 repressive complex
C_2	HOXA13: β -catenin complex
X_{8r}	RA:RAR transcriptional complex
X_{12r}	CYP26A1 enzyme

A.2 Representative parameters

Table 2 lists representative parameters used in the coupled model and their sources.

Table 2: Representative model parameters.

Parameter	Description	Source
k_5	Destruction complex turnover rate	Lee et al. (2003) ; Emerick et al. (2017)
K_{16}	β -catenin:TCF binding constant	Emerick et al. (2017)
v_{18}	APC synthesis rate	Estimated
K_r	HOXA13 repression threshold (APC)	Estimated
k_{h10}	HOXA13: β -catenin association	Estimated
k_{-h10}	HOXA13: β -catenin dissociation	Estimated
γ_0, γ_1	Dynamic K_{16} baseline and gain	Estimated
C_s	HOXA13: β -catenin saturation scale	Estimated
v_{19}	CYP26A1 synthesis rate	Funk (2025)

Parameters not listed here are taken directly from published WNT and RA models or chosen to ensure biologically plausible steady states.

B Nondimensionalization and dimensionless variables

We nondimensionalize time using the destruction-complex turnover rate k_5 (reaction r_5 in the WNT module),

$$\tau = k_5 t,$$

so that one unit of dimensionless time corresponds to the characteristic deactivation timescale of the destruction complex,

$$\frac{1}{k_5} = \frac{1}{0.133} \approx 7.5 \text{ min.}$$

All concentrations are rescaled by a common reference level $\sigma = 1000 \text{ nM}$, i.e.,

$$X(t) = \sigma x(\tau),$$

with σ chosen from the largest steady-state concentrations in the RA subsystem to provide a consistent magnitude scale across WNT, RA, and HOX variables.

After rescaling, the coupled system can be written as

$$\frac{d\mathbf{z}}{d\tau} = \mathbf{F}(\mathbf{z}; \boldsymbol{\theta}),$$

where $\mathbf{z}$ collects all dimensionless state variables and $\boldsymbol{\theta}$ denotes the dimensionless parameter set.

C Full subsystem equations (dimensional form)

C.1 Full WNT/ β -catenin signaling system

The WNT/ β -catenin subsystem follows the reaction-kinetic framework introduced by [Lee et al. \(2003\)](#) and extended by [Emerick et al. \(2017\)](#). It consists of 17 state variables and 21 reactions describing destruction-complex assembly, β -catenin phosphorylation, degradation, stabilization, and transcriptional activation.

We retain the original notation $X_1, \dots, X_{17}$ and reaction rates $r_1, \dots, r_{21}$. The governing equations are written in stoichiometric form

$$\dot{\mathbf{X}} = S \mathbf{r}(\mathbf{X}),$$

where S is the stoichiometric matrix and $\mathbf{r}$ the vector of reaction rates.

All reactions, rate laws, and parameter values not explicitly modified by HOX or RA coupling are identical to those reported in [Emerick et al. \(2017\)](#) and are reproduced below for completeness.

$$\frac{dX_2}{dt} = k_1 W (Dsh^0 - X_2) - k_2 X_2, \quad (23)$$

Table 3: Dimensionless variables and scaling used in the coupled WNT–RA–HOX model. All concentrations are rescaled by $\sigma = 1000$ nM.

Dimensional variable	Dimensionless variable	Scaling	Diagram notation
HOX subsystem			
H_1	H_a	$H_1 = \sigma H_a$	[HOXA5]
M_1	M_i	$M_1 = \sigma M_i$	[MIZ1]
M_2	M	$M_2 = \sigma M$	[MYC]
C_1	C_a	$C_1 = \sigma C_a$	[MYC:MIZ1]
H_2	H_i	$H_2 = \sigma H_i$	[HOXA13]
C_2	C_i	$C_2 = \sigma C_i$	[HOXA13; β -cat]
RA subsystem			
X_{1r}	R_i	$X_{1r} = \sigma R_i$	[Retinol]
X_{2r}	R_a	$X_{2r} = \sigma R_a$	[Retinal]
X_{3r}	A_i	$X_{3r} = \sigma A_i$	[ALDH]
X_{4r}	R	$X_{4r} = \sigma R$	[RA]
X_{5r}	C_b	$X_{5r} = \sigma C_b$	[CRABP _s]
X_{6r}	B_r	$X_{6r} = \sigma B_r$	[RA:CRABP _s]
X_{7r}	A_r	$X_{7r} = \sigma A_r$	[RAR]
X_{8r}	R_r	$X_{8r} = \sigma R_r$	[RA:RAR]
X_{9r}	C	$X_{9r} = \sigma C$	[CYP]
X_{10r}	R_c	$X_{10r} = \sigma R_c$	[RA:CYP]
X_{11r}	R_b	$X_{11r} = \sigma R_b$	[RA:CRABP _s :CYP]
X_{12r}	B_c	$X_{12r} = \sigma B_c$	[RA:CRABP _s :RAR]
X_{13r}	P_a	$X_{13r} = \sigma P_a$	[CRABP _s :CYP]
WNT subsystem			
X_2	V	$X_2 = \sigma V$	[Dsh _a]
X_3	D_a	$X_3 = \sigma D_a$	[APC*:Axin*:GSK]
X_4	D_i	$X_4 = \sigma D_i$	[APC:Axin:GSK]
X_7	P	$X_7 = \sigma P$	[APC]
X_9	D_b	$X_9 = \sigma D_b$	[β -cat*:Axin*:GSK]
X_{10}	B_p	$X_{10} = \sigma B_p$	[β -cat*]
X_{11}	B_a	$X_{11} = \sigma B_a$	[β -cat]
X_{12}	X	$X_{12} = \sigma X$	[Axin]
X_{14}	B_t	$X_{14} = \sigma B_t$	[β -cat:TCF]

$$\begin{aligned}
\frac{dX_4}{dt} = & -(k_3X_2 + k_4 + k_{-6})X_4 + k_5X_3 \\
& + \frac{k_6K_{21}X_7X_{12}}{K_7(K_{21} + X_{12})} \left[GSK^0 - \left(1 + \frac{X_{11}}{K_8} \right) X_3 - X_4 - X_9 \right],
\end{aligned} \tag{24}$$

$$\frac{dX_9}{dt} = \frac{k_9X_3X_{11}}{K_8} - k_{10}X_9, \tag{25}$$

$$\frac{dX_{10}}{dt} = k_{10}X_9 - k_{11}X_{10}, \tag{26}$$

$$\left(1 + \frac{X_{11}}{K_8}\right) \frac{dX_3}{dt} + \frac{X_3}{K_8} \frac{dX_{11}}{dt} = k_4 X_4 - \left(k_5 + \frac{k_9 X_{11}}{K_8}\right) X_3 + k_{10} X_9. \quad (27)$$

$$\begin{aligned} \frac{X_{11}}{K_8} \frac{dX_3}{dt} + \frac{X_{11}}{K_{17}} \frac{dX_7}{dt} + \left(1 + \frac{X_3}{K_8} + \frac{\kappa TCF^0}{(\kappa + X_{11})^2} + \frac{X_7}{K_{17}} + \frac{X_{12}}{K_{20}}\right) \frac{dX_{11}}{dt} \\ + \frac{X_{11}}{K_{20}} \frac{dX_{12}}{dt} = v_{12} + k_{-h10} C_2 \\ - \left(k_{13} + \frac{k_9 X_3}{K_8} + k_{h10} H_2 + k_{h11} H_1\right) X_{11}. \end{aligned} \quad (4)$$

$$\begin{aligned} \left(1 + \frac{X_{11}}{K_8}\right) \frac{dX_3}{dt} + \left(1 + \frac{X_{12}}{K_7} + \frac{X_{11}}{K_{17}}\right) \frac{dX_7}{dt} + \left(\frac{X_3}{K_8} + \frac{X_7}{K_{17}}\right) \frac{dX_{11}}{dt} + \frac{X_7}{K_7} \frac{dX_{12}}{dt} \\ = \frac{v_{18}}{1 + \frac{TCF^0 X_{11}}{K_t(\kappa(C_2) + X_{11})} + \frac{X_{11}}{K_b} + \frac{H_2}{K_r}} \\ - k_{19} X_7 - \left(\frac{dX_4}{dt} + \frac{dX_9}{dt}\right), \end{aligned} \quad (28)$$

$$\begin{aligned} - \frac{X_{12}(K_8 + X_{11})}{K_8(K_{21} + X_{12})} \frac{dX_3}{dt} + \frac{X_{12}}{K_7} \frac{dX_7}{dt} + \left(\frac{X_{12}}{K_{20}} - \frac{X_3 X_{12}}{K_8(K_{21} + X_{12})}\right) \frac{dX_{11}}{dt} \\ + \left\{1 + \frac{X_7}{K_7} + \frac{X_{11}}{K_{20}} + \frac{K_{21} \left[GSK^0 - \left(1 + \frac{X_{11}}{K_8}\right) X_3 - X_4 - X_9\right]}{(K_{21} + X_{12})^2}\right\} \frac{dX_{12}}{dt} \\ = v_{14} + (k_3 X_2 + k_{-6}) X_4 - \frac{k_{15} X_7 X_{12}}{K_m + X_7} + \frac{X_{12}}{K_{21} + X_{12}} \left(\frac{dX_4}{dt} + \frac{dX_9}{dt}\right) \\ - \frac{k_6 K_{21} X_7 X_{12}}{K_7(K_{21} + X_{12})} \left[GSK^0 - \left(1 + \frac{X_{11}}{K_8}\right) X_3 - X_4 - X_9\right], \end{aligned} \quad (29)$$

C.2 Retinoic acid metabolism and signaling system

The RA subsystem is adapted from the retinoid signaling model of [Funk \(2025\)](#), which explicitly tracks retinol uptake, intracellular conversion to retinoic acid, RAR-mediated transcriptional signaling, and CYP26A1-driven degradation.

The subsystem consists of 13 state variables ($X_{1r}, \dots, X_{13r}$) governed by mass-action kinetics and first-order turnover. The full dimensional equations are

$$\frac{dX_{1r}}{dt} = k_{-r1} X_{2r} - k_{r1} X_{1r} + g(t), \quad (30)$$

$$\frac{dX_{2r}}{dt} = k_{r1}X_{1r} - k_{-r1}X_{2r} - k_{r2}X_{2r}X_{3r}, \quad (31)$$

$$\frac{dX_{3r}}{dt} = v_{11} - k_{r23}X_{3r}, \quad (32)$$

$$\begin{aligned} \frac{dX_{4r}}{dt} = & k_{r2}X_{2r}X_{3r} - k_{r3}X_{4r}X_{5r} + (k_{-r3} + k_{r14})X_{6r} - k_{r4}X_{4r}X_{7r} \\ & + (k_{-r4} + k_{r15})X_{8r} - k_{r5}X_{4r}X_{9r} + k_{-r5}X_{10r}, \end{aligned} \quad (33)$$

$$\frac{dX_{5r}}{dt} = v_{16} - k_{r20}X_{5r} - k_{r3}X_{4r}X_{5r} + k_{-r3}X_{6r} + k_{r10}X_{12r}, \quad (34)$$

$$\begin{aligned} \frac{dX_{6r}}{dt} = & k_{r3}X_{4r}X_{5r} - k_{-r3}X_{6r} - k_{r6}X_{6r}X_{9r} + k_{-r6}X_{11r} \\ & - k_{r9}X_{6r}X_{7r} + k_{-r9}X_{12r} - k_{r14}X_{6r}, \end{aligned} \quad (35)$$

$$\frac{dX_{7r}}{dt} = v_{17} - k_{r21}X_{7r} - k_{r4}X_{4r}X_{7r} + k_{-r4}X_{8r} - k_{r9}X_{6r}X_{7r} + k_{-r4}X_{12r}, \quad (36)$$

$$\frac{dX_{8r}}{dt} = k_{r4}X_{4r}X_{7r} - k_{-r4}X_{8r} - k_{r15}X_{8r} + k_{r10}X_{12r}, \quad (37)$$

$$\begin{aligned} \frac{dX_{9r}}{dt} = & v_{19} - k_{r22}X_{9r} - k_{r5}X_{4r}X_{9r} \\ & + (k_{-r5} + k_{r8})X_{10r} - k_{r6}X_{6r}X_{9r} + k_{-r6}X_{11r}, \end{aligned} \quad (38)$$

$$\frac{dX_{10r}}{dt} = k_{r5}X_{4r}X_{9r} - (k_{-r5} + k_{r8})X_{10r}, \quad (39)$$

$$\frac{dX_{11r}}{dt} = k_{r6}X_{6r}X_{9r} - (k_{-r6} + k_{r7})X_{11r}, \quad (40)$$

$$\frac{dX_{12r}}{dt} = k_{r9}X_{6r}X_{7r} - (k_{-r9} + k_{r10})X_{12r}, \quad (41)$$

$$\frac{dX_{13r}}{dt} = k_{r7}X_{11r}. \quad (42)$$

The coupling to the WNT-HOX network occurs through the CYP26A1 synthesis term and the RA-RAR complex X_{8r} , which regulates HOXA5 transcription in the main model.

C.3 HOX regulatory subsystem

The HOX subsystem encodes cross-regulation between HOXA5, HOXA13, MYC, and MIZ1. The governing equations are

$$\frac{dH_1}{dt} = v_{h1} - k_{h2}H_1, \quad (43)$$

$$\frac{dM_1}{dt} = v_{h3} - k_{h4}M_1 - k_{h5}M_2M_1 + k_{-h5}C_1, \quad (44)$$

$$\frac{dM_2}{dt} = v_{h6} - k_{h7}M_2 - k_{h5}M_2M_1 + k_{-h5}C_1, \quad (45)$$

$$\frac{dC_1}{dt} = k_{h5}M_2M_1 - k_{-h5}C_1, \quad (46)$$

$$\frac{dH_2}{dt} = v_{h8} - k_{h9}H_2 - k_{h10}H_2X_{11} + k_{-h10}C_2, \quad (47)$$

$$\frac{dC_2}{dt} = k_{h10}H_2X_{11} - k_{-h10}C_2. \quad (48)$$

All HOX reactions are modeled using mass-action kinetics, with transcriptional regulation encoded through saturating rate laws.

D Full subsystem equations (dimensionless form)

This section records the full dimensionless subsystems obtained from the scalings in Appendix B. All variables are dimensionless and time is $\tau = k_5 t$.

D.1 Dimensionless WNT/ β -catenin subsystem

For completeness, we reproduce the dimensionless form of the WNT subsystem using the notation introduced in Table 3. The system includes implicit relations inherited from the original reaction-kinetic formulation.

$$\frac{dV}{d\tau} = k'_1 W (dsh^0 - V) - k'_2 V, \quad (49)$$

$$\begin{aligned} \frac{dD_i}{d\tau} = & -(k'_3 V + k'_4 + k'_{-6}) D_i + D_a \\ & + \frac{k'_6 P X}{K_7(K_{21} + \sigma X)} [gsk^0 - (1 + K'_8 B_a) D_a - D_i - D_b], \end{aligned} \quad (50)$$

$$\frac{dD_b}{d\tau} = k'_9 D_a B_a - k'_{10} D_b, \quad (51)$$

$$\frac{dB_p}{d\tau} = k'_{10} D_b - k'_{11} B_p, \quad (52)$$

$$(1 + K'_8 B_a) \frac{dD_a}{d\tau} + K'_8 D_a \frac{dB_a}{d\tau} = k'_4 D_i - (1 + k'_9 B_a) D_a + k'_{10} D_b, \quad (53)$$

$$\begin{aligned} K'_8 B_a \frac{dD_a}{d\tau} + K'_{17} B_a \frac{dP}{d\tau} + \left(1 + K'_8 D_a + \frac{\kappa TCF^0}{(\kappa + \sigma B_a)^2} + K'_{17} P + K'_{20} X \right) \frac{dB_a}{d\tau} \\ + K'_{20} B_a \frac{dX}{d\tau} = v'_{12} + k'_{-h10} C_i - (k'_{13} + k'_9 D_a + k'_{h10} H_i + k'_{h11} H_a) B_a, \end{aligned} \quad (54)$$

$$\begin{aligned} (1 + K'_8 B_a) \frac{dD_a}{d\tau} + (1 + K'_7 X + K'_{17} B_a) \frac{dP}{d\tau} + (K'_8 D_a + K'_{17} P) \frac{dB_a}{d\tau} + K'_7 P \frac{dX}{d\tau} \\ = \frac{v'_{18}}{1 + \frac{B_a}{K'_t(\kappa + \sigma B_a)} + K'_b B_a + K'_r H_i} \\ - k'_{19} P - \left(\frac{dD_i}{d\tau} + \frac{dD_b}{d\tau} \right), \end{aligned} \quad (55)$$

$$\begin{aligned} - \frac{\sigma X(1 + K'_8 B_a)}{(K_{21} + \sigma X)} \frac{dD_a}{d\tau} + K'_7 X \frac{dP}{d\tau} + \left(K'_{20} X - \frac{K'_8 \sigma D_a X}{K_{21} + \sigma X} \right) \frac{dB_a}{d\tau} \\ + \left\{ 1 + K'_7 P + K'_{20} B_a + \frac{\sigma K_{21} [gsk^0 - (1 + K'_8 B_a) D_a - D_i - D_b]}{(K_{21} + \sigma X)^2} \right\} \frac{dX}{d\tau} \\ = v'_{14} + (k'_3 V + k'_{-6}) D_i - \frac{k'_{15} P X}{K_m + \sigma X} + \frac{\sigma X}{K_{21} + \sigma X} \left(\frac{dD_i}{d\tau} + \frac{dD_b}{d\tau} \right) \\ - \frac{k'_6 P X}{K_{21} + \sigma X} [gsk^0 - (1 + K'_8 B_a) D_a - D_i - D_b], \end{aligned} \quad (56)$$

D.2 Dimensionless retinoic acid metabolism and signaling subsystem

Using the dimensionless variables in Table 3, the RA subsystem is

$$\frac{dR_i}{d\tau} = k'_{-1r} R_a - k'_{1r} R_i + G(\tau), \quad (57)$$

$$\frac{dR_a}{d\tau} = k'_{1r} R_i - (k'_{-1r} + k'_{2r} A_i) R_a, \quad (58)$$

$$\frac{dA_i}{d\tau} = v'_{11} - k'_{23r} A_i, \quad (59)$$

$$\begin{aligned} \frac{dR}{d\tau} = & k'_{2r} R_a A_i - (k'_{3r} C_b + k'_{4r} A_r + k'_{5r} C) R + (k'_{-3r} + k'_{14r}) B_r \\ & + (k'_{-4r} + k'_{15r}) R_r + k'_{-5r} R_c, \end{aligned} \quad (60)$$

$$\frac{dC_b}{d\tau} = v'_{16} - (k'_{3r} R + k'_{20r}) C_b + k'_{-3r} B_r + k'_{10r} B_c, \quad (61)$$

$$\frac{dB_r}{d\tau} = k'_{3r} R C_b - (k'_{-3r} + k'_{6r} C + k'_{9r} A_r + k'_{14r}) B_r + k'_{-6r} R_b + k'_{-9r} B_c, \quad (62)$$

$$\frac{dA_r}{d\tau} = v'_{17} - (k'_{21r} + k'_{4r} R + k'_{9r} B_r) A_r + k'_{-4r} R_r + k'_{-9r} B_c, \quad (63)$$

$$\frac{dR_r}{d\tau} = k'_{4r} R A_r - (k'_{-4r} + k'_{15r}) R_r + k'_{10r} B_c, \quad (64)$$

$$\frac{dC}{d\tau} = v'_{19} - (k'_{22r} + k'_{5r} R + k'_{6r} B_r) C + (k'_{-5r} + k'_{8r}) R_c + k'_{-6r} R_b, \quad (65)$$

$$\frac{dR_c}{d\tau} = k'_{5r} R C - (k'_{-5r} + k'_{8r}) R_c, \quad (66)$$

$$\frac{dR_b}{d\tau} = k'_{6r} B_r C - (k'_{-6r} + k'_{7r}) R_b, \quad (67)$$

$$\frac{dB_c}{d\tau} = k'_{9r} B_r A_r - (k'_{-9r} + k'_{10r}) B_c, \quad (68)$$

$$\frac{dP_a}{d\tau} = k'_{7r} R_b. \quad (69)$$

D.3 Dimensionless HOX regulatory subsystem

The dimensionless HOX subsystem is

$$\frac{dH_a}{d\tau} = v'_{h1} + v'_{h2} M_i + v'_{h3} R_r - k'_{h2} H_a, \quad (70)$$

$$\frac{dM_i}{d\tau} = v'_{h4} - k'_{h4} M_i - k'_{h5} M M_i + k'_{-h5} C_a, \quad (71)$$

$$\frac{dM}{d\tau} = v'_{h6} - k'_{h7} M - k'_{h5} M M_i + k'_{-h5} C_a, \quad (72)$$

$$\frac{dC_a}{d\tau} = k'_{h5} M M_i - k'_{-h5} C_a, \quad (73)$$

$$\frac{dH_i}{d\tau} = v'_{h8} - k'_{h9} H_i - k'_{h10} H_i B_a + k'_{-h10} C_i, \quad (74)$$

$$\frac{dC_i}{d\tau} = k'_{h10} H_i B_a - k'_{-h10} C_i. \quad (75)$$

All dimensionless parameters are obtained from the scalings in [Appendix B](#). For transparency, the variable scalings are listed in [Table 3](#).

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